

DOSY NMR as a Tool for Predicting Optimal Conditions for Hydrogel Formation: The Case of a Hyperbranched Polyglycidol Cross-Linked with Boronic Acids

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ABSTRACT: This article demonstrates the utility of DOSY NMR for the determination of the optimal conditions for the efficient covalent, reversible cross-linking of macromolecules in water for hydrogel formation. The studied model system was hyperbranched polyglycidol (HbPGL) containing numerous diol groups in peripheral regions and two types of boronic acids, that is, $\text{B}(\text{OH})_4^-$ and benzene-1,4-boronic diacid, as cross-linking agents. Diffusion coefficient changes of a polymer in solution, under the influence of various concentrations of cross-linking agent and pH, which influences the equilibrium of the reaction between boronic acids and diols, were recorded.

These data are consistent with the rheological properties, namely the $G'_{\max}(\omega)$ of hydrogels prepared under analogous conditions, from more concentrated solutions of HbPGL. This approach appears to be promising as it facilitates avoiding the loss of a large amount of polymer that is necessary for the elaboration of appropriate conditions for network formation in aqueous media. © 2016 Wiley Periodicals, Inc. *J. Polym. Sci., Part B: Polym. Phys.* **2016**, *54*, 2171–2178

KEYWORDS: boronic cross-linker; gels; polymers; NMR spectroscopy; DOSY NMR; rheology

INTRODUCTION The covalent, reversible cross-linking of macromolecules is a well-known process in the formation of transient, (self-) healing materials.¹ Different mechanical properties can be achieved depending on the number of tie-points (cross-links). The determination of the optimal conditions for the formation of a specific cross-linked material is dependent on the bonds generated in the cross-linking reaction, which in turn is sensitive to several physicochemical parameters, and requires the consumption of significant amounts of polymer. A simple spectroscopic method that is capable of identifying the optimal conditions of network formation for semi-dilute solutions would reduce this burden. However, the positive confirmation of cross-linking between macromolecules is not obvious, especially for low-molecular weight compounds, because other reactive functional groups present along the polymer chains could engage in intramolecular bond formation. The possibility of intramolecular cross-linking is especially high for flexible linear polymers² and cannot be excluded for hyperbranched polymers containing flexible branches.³ The formation of intramolecular bonds at the expense of intermolecular cross-linking reduces the

mechanical strength of the polymer network or may even prevent network formation. The most optimal conditions leading to network formation can be estimated based on simple NMR experiments. However, the confirmation of intermolecular cross-linking bonds on a standard NMR spectra is not absolute; signals can correspond to both intermolecular and intramolecular bonds. Nevertheless, bonds generated between individual macromolecules lead to the formation of aggregates in solution. These aggregates are easily identified based on their reduced diffusion coefficients relative to those of the pure polymer, as larger objects move slower. By comparing the diffusion coefficients of a polymer treated with a cross-linking agent with the diffusion rate of the pure polymer, the effectiveness of a cross-linking reaction at a given condition can be estimated. The diffusion coefficients of these objects present in the solution can be measured by Diffusion Ordered NMR (DOSY NMR) spectroscopy. In polymer research, DOSY NMR has applications in the determination of the distribution and molecular weight of polymers.^{4–6} The technique focuses on supramolecular polymer formation⁷ and supramolecular complexes, such as graft copolymers⁸ or micelles from

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triblock copolymers.⁹ The gelation processes have been investigated by Nonappa et al. for low molecular weight compounds in organic solvents.¹⁰

In this article, we demonstrated the use of DOSY NMR for aqueous semi-diluted solutions to predict the optimal conditions for hydrogel formation. We present correlations between the diffusion coefficients of macromolecules based on DOSY NMR measurements with the rheological characteristics of the obtained hydrogels at higher polymer concentration.¹¹

As a model cross-linking reaction, we investigated a boronic diacid with 1,2-diol groups on hyperbranched polyglycidol. Because this reaction is pH dependent,¹² we monitored the influence of pH on the reaction progress in solution and physical characteristics of the produced hydrogels. We applied two types of boronic cross-linkers, that is, benzene-1,4-boronic diacid and inorganic $\text{B}(\text{OH})_4^-$ ions, at various molar ratios of diol functional groups to boronic acid.

EXPERIMENTAL

Materials

Benzene-1,4-diboronic diacid (BDBA), sodium tetraborate decahydrate (borax, $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$), and deuterium solvents (methanol CD_3OD , deuterium oxide D_2O , 40 w/w% solution of sodium deuteroxide NaOD) were purchased from Aldrich and used as received. Glycidol (Aldrich) was dried with 4-Å molecular sieves and freshly distilled under reduced pressure before use. Deionized water with a resistivity of $0.055 \text{ M}\Omega \text{ cm}^{-1}$ was produced with an Adrona Crystal system (Riga, Latvia) and used for hydrogel preparation.

Synthesis of Hyperbranched Polyglycidol (HbPGL)

Hyperbranched polyglycidol was synthesized by anionic polymerization according to the procedure described by Frey.¹³ Briefly, 10% of the hydroxyl groups in 1,1,1-tris(hydroxymethyl)propane (0.0043 mol; 0.5770 g) were converted into alcoholate form in a reaction with sodium hydride (0.0129 mol; 0.031 g). Then, glycidol (0.4 mol; 29.60 g) was added at a rate of 2 mL/h to a reactor equipped with a mechanical stirrer. The reactor was kept in an oil-bath at 95 °C over 14 h in a dry argon atmosphere. The obtained product was dissolved in methanol. Sodium ions at alcoholate sites were exchanged for protons by passing the polymer through a cation-exchange resin (Dowex 50 WX 4). The product was thrice precipitated from methanol in acetone to remove cyclic byproducts, dried under vacuum, and characterized by ^1H NMR, ^{13}C NMR, and GPC methods (chromatograms and NMR spectra are shown in the Supporting Information in the Figures S1–S3).

All experiments were carried out using HbPGL with $M(\text{GPC}) = 6430$; $\bar{M}_w/\bar{M}_n = 1.65$ (see Supporting Information Figure S1); and DB (degree of branching) = 0.61. The fraction of vicinal diol units was estimated on the basis of the ^{13}C NMR spectrum and was $\sim 34.0 \text{ mol } \%$.

Hydrogel Preparation

Hydrogels were prepared in water by adding a solution of cross-linking agent (BDBA or borax) to an aqueous solution of HbPGL. The solution pH levels were adjusted with sodium hydroxide. Hydrogels based on BDBA were prepared at pH from 10 to 13, whereas those prepared from borax were prepared at pH from 11 to 13. The polymer concentration for hydrogel formation was 242.4 mg/mL.

Instrumentation

NMR Spectroscopy

^1H NMR, DOSY ^1H NMR, and ^{11}B NMR spectroscopy were used to investigate hyperbranched polyglycidol interactions with boronic compounds in aqueous solutions at various pH values. The polymer concentration was 66 mg/mL. The solution pH was adjusted with sodium deuteroxide. All measurements were carried out at 295 K on a Bruker Avance III 500 spectrometer. ^1H NMR and ^{13}C NMR (inverse-gated and DEPT spectra) spectroscopy were used for the structural confirmation of the hyperbranched polyglycidol. Spectra were registered in methanol- d_4 . ^{11}B NMR spectra were recorded in simple one-pulse experiments using a zg pulse program from the Bruker Biospin library. The $\pi/2$ pulse and relaxation delay were set to 8.52 μs and 250 ms, respectively. Spectra were processed with a 1 Hz line broadening Lorentzian function, and a backward prediction was applied to remove unwanted ^{11}B background signals.

For the ^1H DOSY measurements, each sample was stabilized at 295 K for at least 5 min before data accumulation, and the ^1H $\pi/2$ pulse length was checked and adjusted carefully for each sample. The standard Bruker pulse program dsteppgp3s was selected for measurements using double stimulated echo for convection compensation, Longitudinal Eddy Current Delay using bipolar gradient pulses for diffusion, and 3 spoil gradients. The gradient pulse (δ) was set to 3.2 ms and the diffusion time (Δ) was set to 350 ms. The gradient spoil pulse was set to 0.6 ms, the eddy current delay was set to 5.0 ms, and the delay for gradient recovery was set to 0.2 ms. The δ and Δ values were checked and adjusted for each sample to obtain appropriate signal attenuation across the DOSY measurement. The DOSY experiments were run in pseudo 2D mode with gradients varied linearly from 5 to 95% in 16 steps with 16 scans per step. Spectra were processed by TopSpin 3.1 software, supplied by the manufacturer. A 1-Hz line broadening Lorentzian function was applied, and each row was phased and baseline corrected before Fourier transformation in the F2 dimension. The diffusion coefficients for the resolved ^1H signals were extracted using the T_1/T_2 analyze module of the TopSpin program.

Size Exclusion Chromatography (SEC)

SEC chromatography was carried out in aqueous solution of NaNO_3 (0.1 wt %) at a flow rate of 1.0 mL min^{-1} using a chromatograph (Knauer K-501 HPLC pump) equipped with a degasser (4-Channel Degasser; K-5004, Knauer), three TSK-GEL columns: G5000 PW_{XL} + 3000 PW_{XL} + 2500 PW_{XL} ($7.8 \times 300 \text{ mm}$; Tosho; 26 °C), an LDC RI detector and a Viscotek

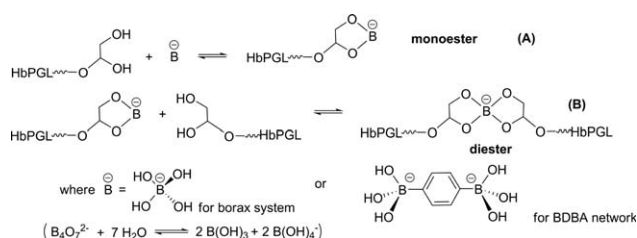


FIGURE 1 Schematic representation of the complexation between HbPGL macromolecules and boronic cross-linkers under basic conditions. In aqueous solution, boronic acid preferentially reacts with diols at a pH above the pKa of boronic acid when boronic acid is in the form of a tetrahedral anion.¹⁵

270 Dual Detector (laser light scattering at $\lambda = 670$ nm (RALS and LALS) and a differential viscometer). The calculation procedure was based on RALS measurements and viscometry and can be found in the literature (i.e., Viscotek manual).¹⁴

Rheology

Rheological measurements were carried out in an Ares rheometer in the viscoelastic regime at 25 °C. The samples were

loaded onto 8-mm diameter parallel plates. The gap distance between the two plates was kept at 1.4 mm. Changes in the storage modulus (G') and loss modulus (G'') were recorded as a function of frequency from 100 to 0.01 rad/s at a 1% strain.

RESULTS AND DISCUSSION

In the examined system of hyperbranched polyglycidol, only the terminal vicinal diol groups in the peripheral areas of the macromolecule were involved in the reaction with boronic acids. The numerous monohydroxyl groups in the linear repeating units remained intact. The reaction of HbPGL with boronic diacid resulted in the formation of cyclic five-membered monoesters and diesters (Fig. 1).

Herein, the formation of a polymer network is only possible due to the diester bonds formed between macromolecules (i.e., intermolecular cross-linking as shown in Fig. 2). As a first example of the benefits of the proposed DOSY NMR technique, the ^{11}B and ^1H NMR spectra (see Supporting Information, Figure S3-S5) of BDBA are not clear and boron spectra could not differentiate between monoesters and diesters. Moreover, while not always unambiguous, a time-consuming deconvolution of signals in the aromatic region

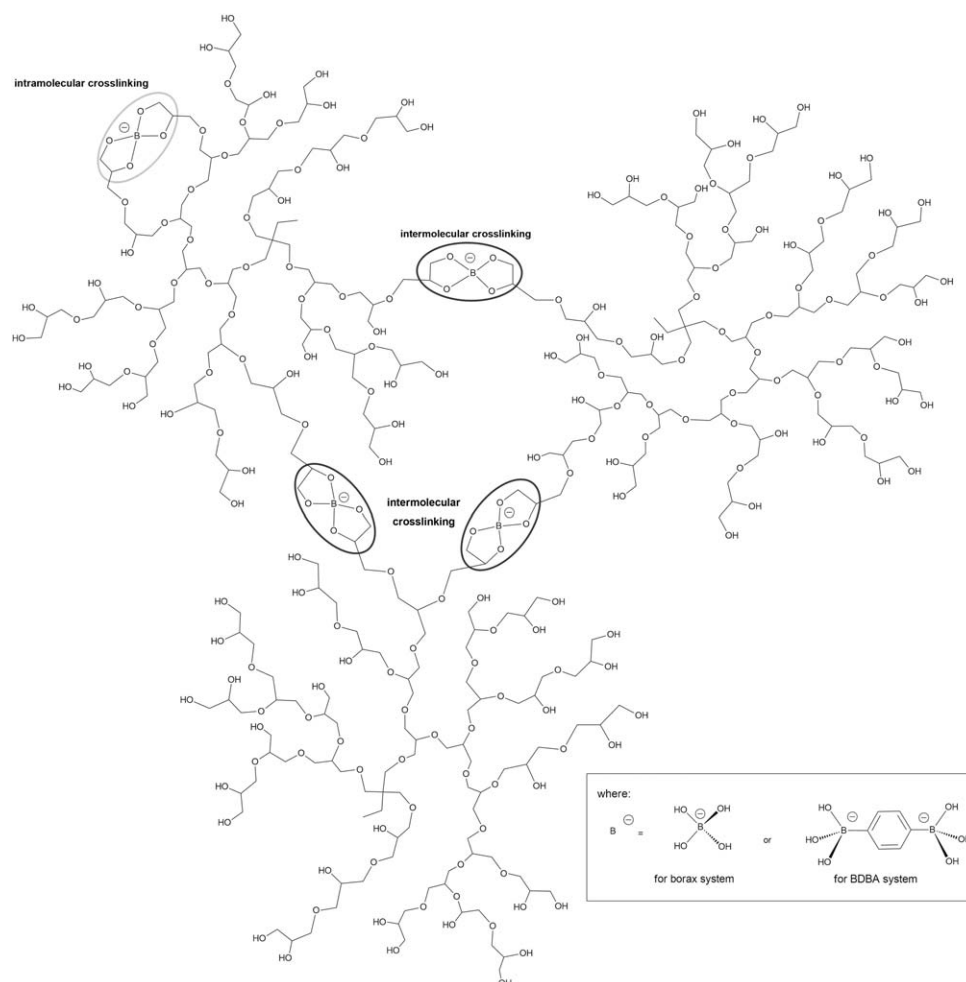


FIGURE 2 Variants of diester cross-links formed within hyperbranched polyglycidol.

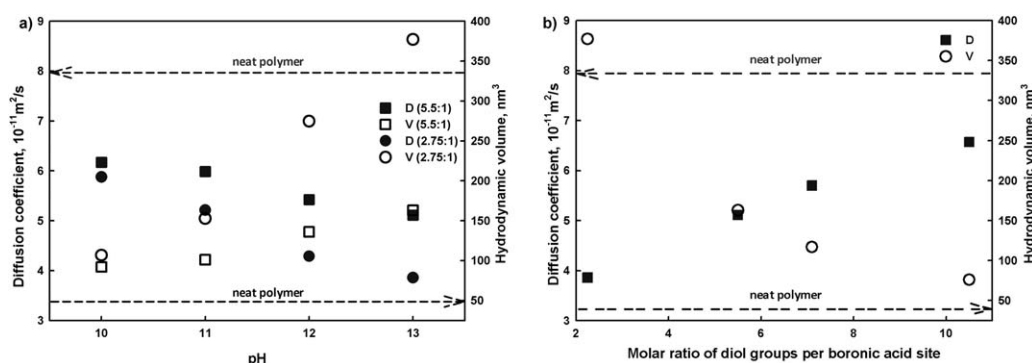


FIGURE 3 Diffusion coefficients (squares) and average molecular volumes (circles) of HbPGL in solution with BDBA as a function of (a) pH at diol to boronic acid site molar ratio of 2.75:1 and 5.50:1 and (b) various diol to boronic acid site molar ratios at pH 13; $C_{\text{HbPGL}} = 66.0 \text{ mg/mL}$.

was necessary for ^1H NMR spectra. Additionally, the presence of signals attributed to specific tie-points (cross-links) does not necessarily indicate network formation; cross-links can be generated between the reactive groups of a single macromolecule (intramolecular bonds, see Fig. 2). A reduced diffusion coefficient corresponding to the characteristic groups of a polymer is a more useful parameter because it indicates increases in the sizes of diffusing objects treated with cross-linking agents. Polymers typically contain several characteristic groups that are visible in NMR spectra that can be used to monitor changes in their diffusion coefficients. However, we must emphasize that the estimation of the cross-linking efficiency using this DOSY NMR technique requires that the bonding interaction is the only factor responsible for cross-linking bond formation and that no other mechanisms of aggregation are present. The comparison of the diffusion coefficients of the pure polymer and of the polymer treated with cross-linking agents facilitates the determination of favorable conditions for the formation of a hydrogel and thereby minimizes polymer consumption.

We were able to confirm the cross-linking of HbPGL macromolecules into larger aggregates by comparing the reduced diffusion coefficients of methine and methylene groups on the repeating units of the polymer (broad signal around $\delta = 3.5 \text{ ppm}$; ^1H NMR spectrum of HbPGL is presented in Supporting Information, Figure S2) with the diffusion coefficients for pure HbPGL. These groups were arbitrarily selected; however, any functional group of the polymer (such as an initiator fragment) could have been considered to predict network formation. The value of the diffusion coefficient represents the average state of the macromolecules, that is, in a freely soluble form and in a polymer network. The conjugation of HbPGL with boronic diacids resulted in a lower diffusion coefficient because the translational mobility of an aggregate is lower than that of a free macromolecule. Furthermore, the diffusion coefficient corresponding to the polymer is identical to the diffusion coefficient determined for the BDBA diester (the signal in the ^1H NMR spectrum at 7.2 ppm).

The hydrodynamic diameters (d_H) of the polymers in solution and, consequently, their molecular volumes were

calculated using the Stokes-Einstein equation (eq 1),^{16,17} where η is the viscosity of deuterium oxide diffusion medium and D is the polymer diffusion coefficient (assuming spherical polymers):

$$\text{Equation } d_H = \frac{kT}{3\pi\eta D} \quad (1)$$

A single macromolecule of HbPGL can be approximated as a sphere-shaped nanoparticle coated with peripheral groups that are reactive toward boronic acids. The hydrodynamic diameter of a pure HbPGL macromolecule was calculated to be 4.36 nm ($D = 7.92 \cdot 10^{-11} \text{ m}^2/\text{s}$), which was consistent with the value obtained from GPC analysis (3.84 nm) using a RALLS detector. The corresponding volume of a single HbPGL macromolecule was 43.4 nm^3 .

Because the reaction equilibrium of the formation of cyclic boronic esters is strongly dependent on the solution pH,¹² we demonstrated the effects of pH on the molecular size of HbPGL aggregates [Fig. 3(a)]. For the reactions with benzene-1,4-boronic diacid, these effects were readily apparent. At an 11.0:1 molar ratio of diol groups to boronic diacid, that is, 5.5 diol groups per reactive boronic acid site, we analyzed the effects at pHs from 10 to 13 (above the pK_a of the boronic acid; Table 1). For DOSY NMR investigations, the weight concentration of the polymer was kept constant at 66 mg/mL. The diffusion coefficient values varied from $6.17 \cdot 10^{-11}$ at pH 10 to $5.11 \cdot 10^{-11} \text{ m}^2/\text{s}$ at pH 13. Consequently, the average molecular volume of the diffusing objects gradually increased from 92 nm^3 at pH 10 reaching 163 nm^3 at pH 13. Additionally, when the molar ratio of the diol functional groups per boronic acid site was 2.75:1, the diffusion coefficient changed from $5.87 \cdot 10^{-11} \text{ m}^2/\text{s}$ at pH 10 to $3.86 \cdot 10^{-11} \text{ m}^2/\text{s}$ at pH 13, which corresponds to aggregate volumes of ~ 107 and 377 nm^3 , respectively. These results indicate that the hydrogel formed at pH 13 would have the highest cross-linking density and the best mechanical strength. At this pH, we also examined the influence of the BDBA concentration on the efficiency of cross-linking. Our studies showed that the BDBA concentration had significant effects on the sizes of diffusing objects. Generally, the

TABLE 1 Diffusion Coefficients, Average Hydrodynamic Diameter (d_H), and Average Molecular Volume (V) of HbPGL Objects in Solution with Benzene-1,4-boronic diacid

pH	Molar ratio of diol to BDBA (diol per one boronic acid site)	D (10^{-11} m ² /s)	d_H (nm)	V (nm ³)
10	2.75:1	5.877	5.89	106.7
	5.50:1	6.170	5.60	92.0
11	2.75:1	5.214	6.63	152.8
	5.50:1	5.987	5.78	101.0
12	2.75:1	4.288	8.07	274.6
	5.50:1	5.420	6.38	136.0
13	2.75:1	3.860	8.96	377.0
	5.50:1	5.111	6.77	163.0
	7.10:1	5.70	6.07	117.0
	10.50:1	6.57	5.26	76.2

trend was as follows: the lower the concentration of BDBA, the smaller the molecular sizes of the aggregates [Fig. 3(b)]. For example, in a sample with the highest cross-linker concentration (i.e., 2.75 mole diol groups per boronic acid site), the diffusion coefficient was significantly reduced, compared with that of the pure polymer, to $3.86 \cdot 10^{-11}$ m²/s, which corresponded to an average hydrodynamic volume of 377 nm³ diffusing objects (hydrodynamic diameter = 8.96 nm). When the polymer and BDBA were mixed at a diol group to boronic acid site molar ratio of 10.5:1, the diffusion coefficient was $6.57 \cdot 10^{-11}$ m²/s, and the aggregate volume was only 76 nm³ (d_H = 5.26 nm). The increased diameters of the diffusing objects resulted from the increased number of diester connections formed per macromolecule, which is due entirely to the increased amount of BDBA. Based on a detailed analysis of the ¹H NMR spectra, after signal deconvolution, the absolute concentration of boronic diester was calculated and the number of cross-links per HbPGL macromolecule was estimated. For example, when the molar ratio of diol to boronic acid site was 2.75, more than three cross-links per HbPGL macromolecule were indicated, and the volume of each polymer aggregate was 377 nm³. When the molar ratio was 5.50, the hydrodynamic volume of each diffusing object was 163 nm³, and the estimated number of cross-links per macromolecule was two. Additionally, for the initial 10.5 molar ratio, one boronic cross-link was generated per macromolecule, and the hydrodynamic volume of each polymer aggregate was only 76 nm³. This can be attributed to HbPGL dimers. The increase in the polymer aggregate size with increasing boronic diester concentration indicates that for specific polymer concentrations selected for DOSY NMR measurements, intermolecular cross-linking plays a predominant role.

Generally, the impact of the cross-linker concentration was more noticeable than the effect of pH. However, in the case of boronic acid/diols complexation, there exists a threshold pH below, which the reaction equilibrium is unfavorable regardless of the amount of reagents used.

The rheological characteristics (based on dynamic viscoelastic measurements) of the hydrogels prepared under corresponding DOSY NMR conditions (i.e., the same pH and diol to cross-linker molar ratios but higher concentration at 242.4 mg/mL of polymer) confirmed the validity of the DOSY NMR results in predicting the most and least favorable conditions of gelation. $G'(\omega) > G''(\omega)$ at high angular frequencies confirmed network formation in water (see Supporting Information Figure S7). The plateau modulus $G'_{\max}(\omega)$ at high frequencies, which is proportional to the total number of association sites, gradually increased with pH and exceeded $6 \cdot 10^5$ Pa at pH 12 and 13. Under these conditions, the largest aggregates were observed with DOSY NMR. Consequently, the experimental number density of elastically effective network chains ($N_{E,\text{exp}}$) also increased¹⁸ [Fig. 4(a)]. The differences in storage modulus maximum (in the plateau region) for hydrogels prepared at various pH were not significant (e.g., $1 \cdot 10^5$ Pa at pH 10 to around $7 \cdot 10^5$ Pa at pH 13). However, with increasing pH, the angular frequency (i.e., the width of the plateau)¹⁹ significantly shifted at the G' and G'' crossover (where G'' was at a maximum) to a lower value, that is, from ω_c (pH 10) = 99.5 rad/s to ω_c (pH 13) = 2.2 rad/s (Fig. 5). The crossover point corresponded to the gelation point, that is, having both a liquid and solid-like behavior. For transient networks, this parameter describes the lifetime (τ) of the gel forming bonds ($\omega_c = 1/2\pi \cdot \tau$).¹⁹ In our case, this parameter is highly pH-dependent. Lower values of ω_c correspond to longer bond lifetimes. Rheological experiments showed that network formation occurred over the entire pH range at room temperature and the applied polymer weight fraction. However, at pH 10 and 11, liquid-like behavior was predominant, whereas at pH 12 and 13, gels were more solid-like. The $G'_{\max}(\omega)$ of the hydrogels prepared at pH 12 and 13 were similar.

At pH 13, we prepared a series of hydrogels with different concentrations of BDBA. An increased molar concentration of BDBA resulted in a significantly increased $G'_{\max}$ [Fig. 4(b)] due to an increased cross-linking density between individual

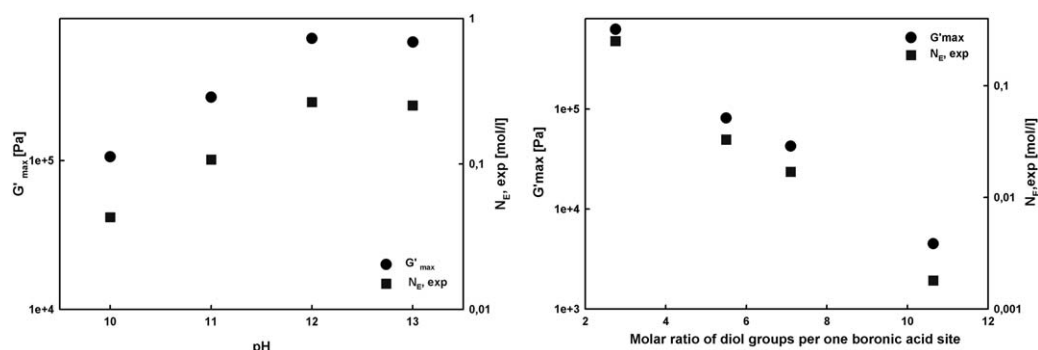


FIGURE 4 Maximum storage modulus, $G'_{\max}(\omega)$ (circles) and the experimental number density of elastically effective network chains, $N_{E, \exp}$ (squares), of hydrogels as a function of (a) pH at diol to boronic acid site molar ratio of 2.75:1; (b) various diol to boronic acid site molar ratios at pH 13; $C_{\text{HbPGL}} = 242.4$ mg/mL.

macromolecules. These observations agreed with the DOSY NMR results. The $G'_{\max}(\omega)$ for the hydrogel at the lowest cross-linker concentration (10.6 diol groups per boronic acid site) was equal $4.5 \cdot 10^3$ Pa, whereas for a diol to acid molar ratio of 2.75:1, the $G'_{\max}(\omega)$ exceeded 10^6 Pa. We expected this trend because a similar trend was observed for semi-diluted solutions where the diffusion coefficient of HbPGL varied from 6.57 to $3.86 \cdot 10^{-11}$ m²/s, respectively. The ω_c value remained constant for all hydrogel systems prepared at pH 13 at $\sim \omega_c = 2.5$ rad/s and did not change with cross-linker concentration at constant polymer fraction and pH.¹⁸

These trends can be used to predict, to some extent, the rheological properties of hydrogels prepared under defined set of conditions.

The consistency between the DOSY NMR and rheological data for HbPGL with BDBA as a cross-linking agent encouraged us to validate our approach by testing borax, a well-known inorganic boronic compound that is used for cross-linking of poly(vinyl alcohol).^{20–23} Borax in aqueous solution undergoes dissociation to $\text{B}(\text{OH})_4^-$ and $\text{B}(\text{OH})_3$ species.

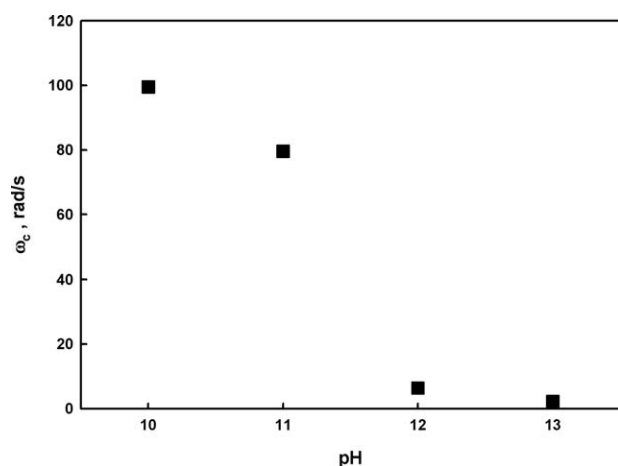


FIGURE 5 Angular frequency values at the crossover points of $G'(\omega)$ and $G''(\omega)$ as determined in frequency sweep measurements carried out for hydrogels prepared at various pHs composed of HbPGL cross-linked with BDBA at 5.5:1 molar ratio.

The interactions efficiency of hyperbranched polyglycidol macromolecules with borax was not comparable with that of benzene-1,4-boronic diacid. Regardless of the concentration of borax used over a pH range of 11 to 13, no distinct increase in the dimensions of the diffusing objects was visible from the DOSY NMR measurements (see Table 2). Polymer objects generated at pH 12 and 13 resulted in volumes twice those of non-cross-linked HbPGL macromolecules. The average hydrodynamic diameters were 5.5 and 5.1 nm at diol to boronic acid site molar ratios 2.75 and 5.50, respectively. The hydrodynamic diameters of objects observed at pH 11 were close to the dimensions of the pure polymer.

The DOSY NMR data for the borax-treated polymers were interpreted for network formation capabilities. The formation of HbPGL hydrogels via cross-linking with $\text{B}(\text{OH})_4^-$ ions was not as favorable as with BDBA. Hydrogels failed to form at pH 11 over the entire range of investigated borax concentrations (at 23 wt % polymer). These results confirmed the DOSY NMR measurements and showed that pH 11 was not favorable for HbPGL interactions with $\text{B}(\text{OH})_4^-$. Similarly, at pH 12 and 13, for diol to $\text{B}(\text{OH})_4^-$ molar ratio of 11.0:1 (5.5:1), only viscous liquids were formed with average hydrodynamic volumes of 71.1 and 71.9, respectively. Hydrogels were obtained at these pH levels when higher concentrations of borax were used, i.e., at molar ratios of 7.0:1 and 5.5:1 (molecular volumes >80 nm³). These results were expected based on the polymer diffusion coefficients. The

TABLE 2 Diffusion Coefficients and Average Hydrodynamic Volume (V) of HbPGL Objects in Solution with Borax

Molar ratio of diol groups to $\text{B}(\text{OH})_4^-$	pH					
	Diffusion coefficient (10^{-11} m ² /s)			Volume (nm ³)		
	11	12	13	11	12	13
5.50:1	7.70	6.27	6.37	47.4	88.0	84.3
7.00:1	7.52	6.46	6.42	50.3	79.7	82.0
11.00:1	6.96	6.73	6.70	63.9	71.1	71.9

TABLE 3 Rheological Parameters of Hydrogels Composed of HbPGL Cross-Linked with Borax at pH 12 at Room Temperature

Molar ratio of diol:B(OH) ₄ [⊖] [diol:0.5B(OH) ₄ [⊖]]	$G'_{\max}$ (Pa)	$N_{E,\text{exp}}$ (mol/l)	ω_c (rad/s)
5.50:1 (2.75:1)	$3.09 \cdot 10^5$	0.1255	2.84
7.00:1 (3.50:1)	$2.30 \cdot 10^5$	0.0974	2.45

characteristics of hydrogels prepared at pH 12 are shown in Table 3.

Generally, hydrogels obtained by cross-linking HbPGL with BDBA displayed better mechanical properties. For example, the hydrogels prepared at pH 12 and a diol to cross-linker molar ratio of 5.50:1 (2.75 diol functional groups per boronic acid site), the plateau value of the storage modulus for BDBA was $6.96 \cdot 10^5$ Pa ($D = 5.42 \cdot 10^{-11}$ m²/s), whereas for the borax system, the storage modulus was only $3.09 \cdot 10^5$ Pa ($D = 6.27 \cdot 10^{-11}$ m²/s). Such a cross-linking difference was unexpected because the pK_a of both boronic acids are comparable. According to the literature, the acidity of boronic acid has a significant effect on the yield of boronic ester formation.²⁴ We suspected higher rates of intramolecular cross-linking with B(OH)₄[⊖] ions. However, these type of interactions may not be as prevalent as indicated by low concentrations of boronic diester species in the NMR spectra of boron ($\delta = 9.80$ ppm, see Supporting Information, Figure S6). For example, for diol to B(OH)₄[⊖] molar ratio of 11.0:1, only one diester junction for three macromolecules of HbPGL was indicated. Under these conditions, the formation of the associated of macromolecules aggregates was impossible. This was consistent with the diffusion coefficient value similar to that of the pure polymer. When the diol to B(OH)₄[⊖] molar ratios were 5.5:1 and 7.0:1, one diester bond corresponded to less than two macromolecules. This indicated that the polymer aggregates primarily formed as HbPGL dimers. The DOSY NMR measurements indicated that objects of average molecular volumes of ~ 80 nm³ corresponded to two cross-linked HbPGL macromolecule aggregates. The calculation of the number of particular boronic species based on the solid-state ¹¹B NMR analysis of hydrogel systems (see Supporting Information Figure S8) for an initial diol to B(OH)₄[⊖] molar ratio of 5.5:1 revealed that the molar ratio of boronic cross-linking bonds (boronic diesters) corresponding to HbPGL macromolecules was ~ 1.7 . However, because the number of cross-linking bonds per macromolecule is not significant and only determines network formation, we were more confident that intramolecular interactions were not favored within the analyzed system.

For comparison, in the BDBA system at pH 13, for a diol to BDBA molar ratio of 11.0:1, two diester bonds were formed per macromolecule. Additionally, an increase in the initial concentration of BDBA (5.5:1) resulted in the formation of more than three diesters per macromolecule. Overall, the DOSY NMR and rheological data revealed that BDBA was a more efficient cross-linker of HbPGL than borax. The

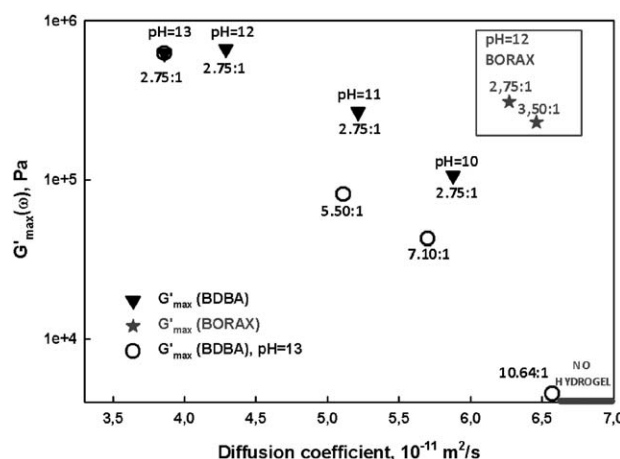
difference of boronic acids reactivity in HbPGL cross-linking can result from different sizes of cross-linker molecules and different number of boron atoms participating in the reaction (in the case of B(OH)₄[⊖] one boron atom participates in the diester formation, whereas in the BDBA two isolated boron atoms are engaged in the formation of boronic diester).

These studies on hyperbranched polyglycidol ($M_n = 6430$) cross-linked with BDBA and borax demonstrated that hydrogel formation for more concentrated systems (23 wt %) was possible, if the DOSY NMR results for semi-dilute solutions (6.25 wt %) indicated the formation of particles with average hydrodynamic volumes above 75.0 nm³ corresponding with diffusion coefficients below $6.60 \cdot 10^{-11}$ m²/s; Fig. 6).

To some extent, these studies demonstrate that we can choose suitable conditions for network formation to obtain hydrogels with specific properties [e.g., $G'_{\max}(\omega)$] based on the relationship between the maximum storage modulus, the crossover frequency of cross-linked material and the diffusion coefficient of polymer objects in a semi-dilute solution. While our approach is applicable for various polymer systems, each tested system would require optimization.

First, the concentration of polymer for the DOSY NMR measurements must be adjusted to a level that enables the observation of size changes of polymer aggregates upon cross-linking. The selected polymer concentration should be in a semi-dilute regime within which macromolecules can be bound via cross-linking agents. The results of DOSY NMR measurements carried out within this polymer concentration regime can then be correlated to the properties of more concentrated systems, for example, the rheological properties of hydrogels. Too dilute polymer concentrations are not suitable because intramolecular cross-linking bonds would dominate due to the relative isolation of individual macromolecules in solution.

As for the other examined variables, such as the cross-linker concentration and pH, to predict the rheological properties of a

**FIGURE 6** $G'_{\max}(\omega)$ values of hydrogels versus diffusion coefficients of solutions containing polymers treated with a cross-linker at various pHs and a diol to boronic acid site molar ratio of 2.75:1.

system based on DOSY NMR measurements, corresponding NMR and rheology measurements should be made at least for the boundary conditions of the system. Moreover, based on changes in material properties obtained by varying one parameter at these boundary conditions data, for example, the pH or cross-linker concentration, we can predict the approximate composition of a mixture necessary to obtain a hydrogel with specific properties. The highest cross-linking agent concentration is usually determined by the concentration of accessible reactive sites in a polymer, whereas the lowest concentration is the value at which gel formation can occur. However, the DOSY NMR measurements can also be used to show that gels cannot be formed for more concentrated systems. Based on the known threshold value of diffusion coefficients of polymers above which networks do not form, the conditions that do not favor cross-linking can easily be rejected. However, the conditions under which DOSY NMR measurements are made should be properly monitored to avoid significant increases in viscosity in the NMR analysis, which can affect the accuracy of the analysis.

CONCLUSIONS

Our research on hyperbranched polyglycidol molecules cross-linked with benzene-1,4-boronic diacid or borax demonstrated the correlation between DOSY NMR measurements of semi-dilute solutions and the rheological characteristics of hydrogels prepared from more concentrated systems. Reduced diffusion coefficients indicated increased $G'_{\max}(\omega)$ plateau values. Moreover, for the hyperbranched polyglycidol system, we were able to determine the threshold value of a diffusion coefficient ($6.60 \cdot 10^{-11} \text{ m}^2/\text{s}$) above which no hydrogel formation occurred for concentrated mixtures. Because this DOSY NMR technique is a reliable method for determining the formation of polymer aggregates under polymer treating with a cross-linking agent, tedious analyses of NMR spectra can be avoided. Moreover, the presence of covalent cross-links in NMR spectra is not the absolute indication of intermolecular cross-linking. Nonetheless, the combination of NMR and DOSY NMR techniques was able to indicate the conditions under which the formation of intramolecular cross-links within a single HbPGL macromolecule were not favorable. Moreover, the different cross-linking abilities of different types of boronic acids was explored using DOSY NMR and rheological analyses.

We demonstrated that the DOSY NMR results could capably predict hydrogel formation and confirmed these predictions via rheological measurements. Our approach seems to be especially suited for novel hydrogel systems based on reactions sensitive to various parameters, such as pH. This approach is promising and can reduce polymer waste in determining the optimal conditions for network formation. These DOSY NMR measurements can be performed using semi-dilute solutions and can determine the optimal conditions for network formation in more concentrated systems. Based on the known trends for $G'_{\max}(\omega)$ under given conditions, we can thus adjust the properties of a prepared hydrogel for specific applications.

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